The Gettier Case File: Knowledge as an Earned Verdict

v1.1 — 2026-10-26. English edition. Version DOI 10.5281/zenodo.23663277. Supersedes v1.0 (2026-08-10, DOI 10.5281/zenodo.21877469). *Changes in v1.1:* Z is named for what it is — a **dirty** parameter, following ZTL v2.1.3 (DOI 10.5281/zenodo.23648237). §2.1 is rewritten: v1.0 called Z an "unverified" tag, one of three statuses of the data, which suggested the wrong reading — one hidden value merely not yet known. The glossary and the phrases that name a Z-marked link follow. No verdict, carrier or measurement changes.

Vitaly Reznik. AI participation disclosed — see Acknowledgements.

Abstract. For twenty-four centuries knowledge was defined as justified true belief (JTB). In 1963 Edmund Gettier showed, in three pages, that a belief can be justified, true — and still not knowledge. Epistemology has been hunting a "fourth condition" ever since, without agreement. We run both of Gettier's canonical cases through an executable logical instrument — the ZTL judge, which returns not a bare verdict but the pair *verdict + warranty* together with a named list of weak links. The findings: (1) a Gettier case is not a defect of inference but **contraband at admission**: the "justification" of JTB is a T admitted on a witness that testifies about a *different* atom; the logic, handed a poisoned marking, honestly answers "earned" — the fraud happens before it; (2) the case has a **measurable signature**: the set of atoms carrying the verdict under the naive marking and the set carrying it after the world's correction are disjoint — "the truth arrived by a different road than the justification" turns from metaphor into a checkable condition on two sets; (3) the passport office (E18) confirms that no paradox is present — only a conditional refusal with a named culprit, which the naive agent smuggled past. The diagnosis is not new: Artemov gave it in Justification Logic in 2008, and we quote his formulation verbatim. Our contribution is an independent corroboration in a foreign formalism (precisely the convergence of methods Williamson asks formal epistemology to seek), executability (every claim of this paper re-checks with one command), and three deltas along cracks Artemov himself named: a mechanical admission gate instead of post-hoc analysis, a warranty ladder instead of a timeless factivity axiom, and epochs instead of absent time. The jurisdiction boundary is measured and declared: fake-barn cases are *not* caught by the signature — and we show why that refusal is correct rather than a gap.

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1. Prologue: a word with no oversight

Nobody polices the word "justification" in epistemology — just as nobody policed the word "paradox" before the census. The classical definition, descending from Plato: to know X is (1) to believe X, (2) for X to be true, (3) to have justification for the belief. Three conditions — and two millennia of agreement.

In 1963 Edmund Gettier published a three-page paper and broke the definition with two counterexamples. Here they are, one card each.

The sheep. You look at a field and see a white fluffy shape of sheep-like form. You believe: "there is a sheep in the field." You have justification — you can *see* it. And it is true — there is a sheep in the field. But the thing you are looking at is a dog that resembles a sheep; the real sheep grazes behind a hill, out of your sight. The belief is justified, true — and it is not knowledge: truth and justification met by accident.

The coins. The company president told Smith: "Jones will get the job." Smith counted the coins in Jones's pocket — ten. Smith infers: "the man who will get the job has ten coins in his pocket." Justified. But the job unexpectedly goes to Smith himself — who also happens, by chance, to have ten coins. The proposition is true, justified, and Smith believed it — yet nobody would call it knowledge.

Since 1963 philosophers have proposed fourth conditions: no false lemmas (Clark), a causal connection (Goldman), process reliability, indefeasibility (Lehrer–Paxson), safety of the belief (Sosa, Williamson). No single repair has won; Floridi argued none can. The problem is listed as open.

The genre of this paper is the genre of *The Paradox Docket*: we do not solve the problem and we do not propose a new theory of knowledge. We bring the dispute to a measuring instrument — and discover that the instrument hands down verdicts in questions that have been argued in words for sixty years. Every verdict re-checks with a single command (§7).

2. The court and its rules

2.1. ZTL in one paragraph

The instrument is built on ZTL (Zero-Trust Logic), generated by a single principle: **truth is never granted on credit**. An atomic claim is either checked — *earned* (T: a completed verification came out for it) or *refuted* (F) — or **dirty** (Z): a parameter that is not ours; its owner may change it between two reads, so each occurrence is read on its own, and only a verification cleans it, replacing it by a checked value. Z is not "unknown" and not "half-true": it is not one hidden value waiting to be learned. A connective returns T only if it holds under every reading of every dirty occurrence, and F otherwise, so a dirty atom is never itself counted as true. The full axiomatics, 394 machine-checked theorems (Lean 4, empty axiom lists) and the corpus of expeditions are in the canonical preprint: *ZTL — Zero-Trust Logic*, Zenodo, DOI 10.5281/zenodo.21318981.

2.2. The judge: a verdict is a pair

The first instrument is the judge (`ztljudge`). Given a formula and a marking (what is earned, what refuted, what on credit), it returns three things: the **verdict** (true/false/refusal), the **warranty** — the quality of the verdict (the highest grade, *hereditary*, means "no path of verifications can revoke it"; the lowest holds only until the first check), and the **disposition** with named weak links: EARNED; ON CREDIT — true, but riding a dirty link; OPEN — not established; REFUTED. The judge never says "something is wrong with your argument" — it says "your conclusion hangs on the dirty link j."

2.3. Carriers: what the verdict stands on

To the judge we add one measurement — **carriers**: the verified atoms the disposition actually rides on. Remove an atom (return it to Z) — does the disposition change? Then it is load-bearing. Carriers operationalize the question "which grounds are holding this verdict up."

2.4. The passport office

The second instrument is the passport office (`zpassport`, E18), familiar to readers of the *Docket*: it counts the classical solutions of a system of definitions and issues a passport of refusal — PARADOX (refusal forever), INTRINSIC (the verdict is forced), UNDERDETERMINED (a blank awaiting a decree), INPUT (until verification), DOWNSTREAM (refusal inherited; culprits named). In this paper it sits in the acquitting role: its job is to show that the Gettier file contains *no paradox*.

3. Three floors: the case is heard

The formalization is shown in full — that is the instrument's discipline: dispute the encoding, not the verdict. The sheep: `s1` = "what I see is a sheep" (world: F — it is a dog), `s2` = "there is a sheep behind the hill" (world: T, but the agent never looked); belief B = `s1 ∨ s2`. The coins: `jg` = "Jones will get the job" (world: F), `jt` = "Jones has ten coins" (T — Smith counted them), `sg` = "Smith will get the job" (world: T, unknown to Smith), `st` = "Smith has ten coins" (world: T, Smith never thought of them); belief B = `(jg ∧ jt) ∨ (sg ∧ st)`. All numbers below are the output of the reference bench `zgettier.py` (E36), Appendix A.

Floor 1: the naive agent. Smith trusts the president and marks `jg = T`; the sheep-watcher marks `s1 = T` — "I can see it." The judge answers: **EARNED, hereditary** — "grounded; the unverified do not matter." Carriers: `[s1]` for the sheep; `[jg, jt]` for the coins.

This is the central count of the indictment, and we underline it: *the logic does not catch Gettier — and it should not*. The judge is impeccable relative to the marking it was handed. The fraud happened before the court: at the moment the president's word was admitted as a verified fact about the appointment, and the sheep-like silhouette as a verified fact about a sheep. There is a witness — but it testifies about a different atom: the silhouette certifies "I see something sheep-shaped," not "that is a sheep." JTB's "justification" is weaker than EARNED: **a Gettier justification is a T admitted on a witness about something else.**

Floor 2: the agent with customs. The gate rule: a T without a sufficient witness is only Z. The president's word remains evidence about the president's words; `jg` goes to Z (the honestly counted `jt` stays T). The judge answers: **OPEN — not**



Figure 1: **The Admission Customs.** A queue of atoms with witness-suitcases before the court. A genuine witness passes toward the hall marked T; a suitcase holding only a sheep-shaped silhouette — a witness about something else — is turned to the waiting bench marked Z, "until verification." Gettier's fraud happens here, before any logic.

established; verify. Knowledge is never claimed; the Gettier situation dissolves before it forms. An agent with customs has no Gettier problem.

Floor 3: the world corrects. Here the bench found a detail we had not planned: an admitted lie cannot be "re-verified" inside logical time — `verify` acts only on marks, and the temporal machinery refuses to re-mark ground (error code E_TL_GROUND). Expelling an admitted lie is an **epoch event** (§22 of the preprint: the chronology of knowledge ≠ the chronology of validity): not "learning more" but *correcting the admission*. After the correction (`s1:=F`, then `s2:=T`; `jg:=F`, then `sg, st:=T`) the verdict first falls to OPEN, then the belief is **EARNED** again — but the carriers are now `[s2]` for the sheep and `[sg, st]` for the coins.

4. The signature: truth arrived by another road

Fold the three floors into one checkable condition. **The Gettier signature:**

> the belief is EARNED under the naive marking; the naive marking contains an admitted T that the world refutes; the belief is EARNED under the world's marking; and — the carriers of the naive verdict are disjoint from the carriers of the final one:
>
> carriers(naive) ∩ carriers(world) = ∅.

Measured: the sheep — `['s1']` against `['s2']`, empty intersection; the coins — `['jg','jt']` against `['sg','st']`, empty intersection. "Truth and justification met by accident" was a metaphor for sixty years; it is now a condition on two sets of atoms, computed in milliseconds.

The control. Honest knowledge (the sheep is really there and really seen; the world agrees with s1) yields the same EARNED — but the carriers coincide with the world's truth-makers: no signature. The moral: EARNED alone does not separate knowledge from Gettier; EARNED *plus carrier-match* does. A court that cannot acquit is worth nothing; the control is its acquittal.

The coins detail, on its own line. Smith earned `jt` honestly — he *counted* the coins. A genuine, conscientious witness. It does not save him: the load-bearing atom was the poisoned `jg`. One honest verification does not launder one smuggled atom — partial diligence does not convert into knowledge.

The passport axis. Run the honestly-marked sheep through the passport office: `s1 → INPUT`, `s2 → INPUT`, `B → DOWNSTREAM (conditional)` — the refusal is inherited, the culprits named. After the world's correction — everything GROUNDED. And not a single PARADOX cell. In *Docket* terms: **the Gettier file is not a paradox but a conditional refusal with a named culprit, whom the naive agent walked past customs.** The judge and the passport office are two axes of one core, and they agree.

5. The honest frame: who cut this before us

The diagnosis is Artemov's. In Justification Logic, Sergei Artemov formalized Gettier's case back in 2008 (*The Logic of Justification*, §10) and stated the diagnosis we quote verbatim: *"One can see clearly the essence of Gettier's example. In (33), one of two disjuncts is justified but false, whereas the other disjunct is unjustified but true. The resulting disjunction (33) is both justified and true, but not really known to Smith"* (Comment 10.5). More: his *streamlined case* (§10.5 — Artemov showed the coins and pockets to be redundant) coincides structurally with our sheep. We arrived at the same cut independently, in a foreign formalism — and learned of the coincidence during the literature search, as once with Bochvar. This does not diminish; it strengthens: two irreducible machineries — justification terms over classical logic there, a two-valued default-deny valuation here — cut the case in the same place. That is exactly the *convergence of results between methods* that Williamson (2015) prescribes as the cure for distrust of case-intuitions and of single formalizations alike.

What we add to Artemov's diagnosis — three things, each along a crack he named himself:

1. **A gate instead of post-hoc analysis.** In Artemov's setting the agent's justification is an unanalyzed assumption (u:(d)); the contraband enters the formalism as freely as it enters our naive marking, and is caught afterwards, by model analysis. Our gate is a mechanical entry policy: a T without a sufficient witness is not admitted. Prevention instead of diagnosis.



Figure 2: **Two Roads of Truth.** The castle is the belief B. The road of justification — s1, "what I see is a sheep" — breaks off into the void: the bridge is down, the planks float in the river, and the dog he actually saw watches from the far slope. The traveler plants his EARNED flag one step from the gap. The truth reached B by the other road — s2, past the grazing sheep. The carriers do not intersect: truth arrived by a different road than the justification.

2. **A warranty ladder instead of an axiom.** In Artemov's systems factivity is an axiom a logic either adopts (JT) or not; he himself shows (§11) that under factivity Gettier's assumptions are inconsistent. Here factivity is not postulated — it is *earned* per verdict and comes in grades: until-verification / sound / hereditary.
3. **Epochs instead of absent time.** Artemov states plainly (§2.4) that temporal and update readings lie outside his language. Here the discovery of the dog is an epoch event (expire), mechanically distinct from a knowledge tick (verify) — a distinction the bench found on its own, by the E_TL_GROUND refusal.

Four patches — one gate. The classical fourth conditions land on this machinery as four aspects of one admission discipline: "no false lemmas" = no F-atom among the carriers; the causal condition = a witness about the very atom; reliabilism = the quality of the witness; safety/sensitivity = the hereditary grade under refinement. What competes in the literature is, here, four projections of one admission rule.

Floridi, read through this optic. Floridi (2004) argued that the Gettier problem is logically unsolvable: coordinating truth and justification is equivalent to the coordinated-attack (two generals) problem. If he is right, then demanding a fourth condition that *guarantees* the coordination is demanding the impossible. But then the honest response is precisely ZTL's architecture: do not guarantee the coordination — *grade* it (the warranty ladder), forbid the credit (default deny), and admit that no contentful guarantee survives an unrestricted change of epochs (the Epoch Boundary Theorem). Unsolvability is an argument not against knowledge, but against free knowledge.

6. The limits of jurisdiction

A census that does not know its own borders is just one more total theory; we repeat the *Docket*'s gesture and name the edges explicitly — one of them measured inside the bench.

Fake barns (Goldman, 1976). Henry sees a *real* barn — in a county strewn with papier-mâché façades. The witness is honest and points at the true atom; the carriers coincide; the signature **does not fire** — yet epistemologists still deny Henry knowledge. The bench reproduces this non-firing with an assert: the fake-barn failure is not misattributed warrant but **witness sufficiency relative to an environment** ("is a barn-percept a sufficient witness HERE?") — a question addressed to the gate's policy, separate work with separate measurements. Artemov cuts the barns through justification identity (perceptual u versus inferential a·v — §4 of his paper); our gate reframing is complementary. The signature catches misattributed warrant, not unreliable environments — and we exhibit this as a measured refusal, not as a gap.

Outside the jurisdiction entirely: the lottery paradox (probability), the sorites (vagueness), the knowability paradox (second-order modality), experimental philosophy (whose intuitions count). Some of these families are covered by other instruments of the ZTL corpus; some by no one.

7. Don't believe it — press the button

Two ways to recompute this paper.

Locally: the repository https://github.com/inventor1975/ZTL,

```
python3 zgettier.py
```

— the reference run of E36: both cases, the control, the jurisdiction limit, the passport axis; every claim of this paper is pinned by an assert; a failing assert is a refutation of this paper. The invitation to falsify comes with the genre.

In the browser: ZTLStudio — https://ztl.vitalyreznik.com. In the assertion genre: mark `s1=T`, ask `s1 | s2` — receive a verdict with its warranty; drop the marking — see OPEN and the weak links; in the system genre: assemble the sheep and receive the INPUT/DOWNSTREAM passports. The built-in AI only translates — the core judges.

8. Conclusion: the price of the word "justification"

What has changed after the hearing? The word "justification," which epistemology used without oversight, has received a price list. Justification-as-in-Gettier is credit: a T admitted on a witness about a different atom; a logic handed such a marking goes honestly bankrupt together with it. Knowledge is not JTB and not JTB+X: it is **EARNED over an admissible marking** — a truth every carrier of which was earned by a witness about that very atom. What Gettier sold philosophy was neither a paradox nor a mystery — it was a window onto the customs desk over which, for sixty years,

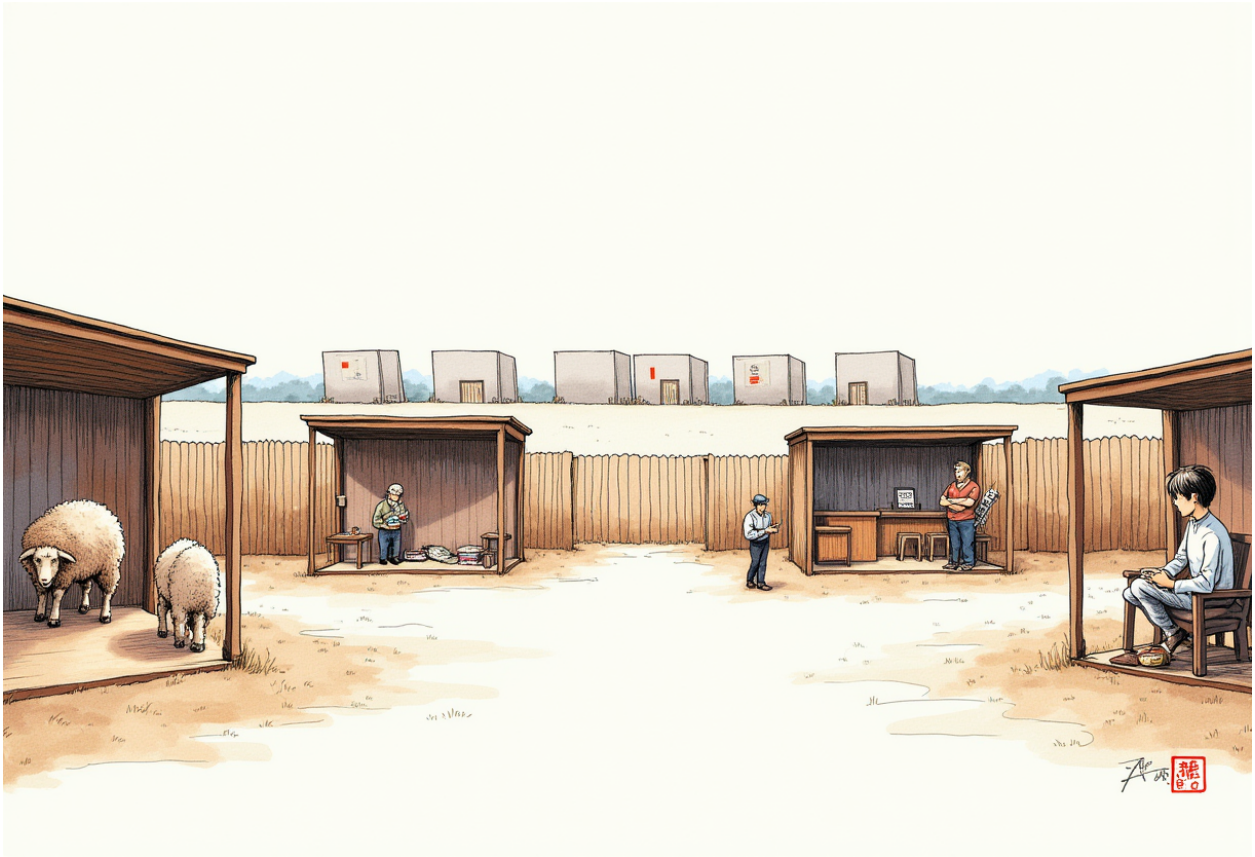


Figure 3: **The Group Photo, by Axes.** Left enclosure — honest knowledge: the shepherd with a real sheep (EARNED, carriers match). Middle — the Gettier pen: the clerk with his coins (EARNED, carriers disjoint). Right — the gated agent, calmly waiting (OPEN, knowledge not claimed). Beyond the fence, a different jurisdiction: Henry before a row of cardboard barn façades.

hung the sign: “T carried through without inspection.” The inspection takes milliseconds. The only expensive thing was the time lived without it.

Acknowledgements and disclosure

This work was carried out with the substantial participation of the AI system Claude Fable 5 (Anthropic) in a dialogue setting (the corrections of v1.1 — the dirty parameter — with Claude Opus 5.5, Anthropic), under the Variant A collaboration model: the human — in this project’s vocabulary, the curator — sets the tasks, verifies the results, and answers for the content. In accordance with COPE/ICMJE recommendations, the AI system is not listed as an author; the human author bears full responsibility. The reliability of the results does not depend on trusting the AI: every claim re-checks via the repository code. Instruments: ZTL (concept DOI 10.5281/zenodo.21318981), the modules `ztljudge` and `zpassport` (E18), the reference bench `zgettier.py` (E36), and the ZTLStudio web interface.

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Appendix A. The reference run

The full output of `python3 zgettier.py`, identical to the one published in the repository at the publication commit:

```
============================================================================
E36: THE GETTIER BENCH — knowledge as an earned verdict
============================================================================
----------------------------------------------------------------------------
CASE I. THE SHEEP IN THE FIELD (Chisholm's costume of Case I)
  s1 = 'what I see in the field is a sheep'   world: F (a dog)
  s2 = 'there is a sheep behind the hill'     world: T (unseen)
  belief B = s1 | s2   ('there is a sheep in the field')
  [floor 1: naive agent, s1 admitted as T]
    (s1 ∨ s2)   marking={'s1': 'T'}
    -> EARNED (verdict T, grade hereditary)   carriers=['s1']
  [floor 2: gated agent, unwitnessed T -> Z]
    (s1 ∨ s2)   marking={}
    -> OPEN (verdict F, grade until-verification)   carriers=[]
  [floor 3: world corrects (epoch): s1:=F, s2:=T]
    (s1 ∨ s2)   marking={'s1': 'F', 's2': 'T'}
    -> EARNED (verdict T, grade hereditary)   carriers=['s2']
```

```
  GETTIER SIGNATURE: carriers ['s1'] vs ['s2'], intersection = [] ✓
  passport axis: s1,s2=INPUT, B=DOWNSTREAM(conditional); after the
  correction all GROUNDED; zero PARADOX — Gettier is not a paradox,
  it is a conditional refusal whose culprit the naive agent smuggled past.
------------------------------------------------------------------------
CASE II. GETTIER'S OWN TEN COINS (Case I of the 1963 paper)
  jg = 'Jones will get the job'    world: F (the boss was wrong)
  jt = 'Jones has ten coins'       world: T (Smith COUNTED them)
  sg = 'Smith will get the job'    world: T (unknown to Smith)
  st = 'Smith has ten coins'       world: T (Smith never thought of it)
  belief B = (jg & jt) | (sg & st)   ('the man who will get the job
  has ten coins in his pocket')
  [floor 1: naive Smith (boss's word admitted as T)]
    ((jg ∧ jt) ∨ (sg ∧ st))   marking={'jg': 'T', 'jt': 'T'}
    -> EARNED (verdict T, grade hereditary)   carriers=['jg', 'jt']
  [floor 2: gate (boss's word is not an appointment), jt stays T]
    ((jg ∧ jt) ∨ (sg ∧ st))   marking={'jt': 'T'}
    -> OPEN (verdict F, grade until-verification)   carriers=[]
  [floor 3: world corrects: jg:=F; sg,st verified T]
    ((jg ∧ jt) ∨ (sg ∧ st))   marking={'jg': 'F', 'jt': 'T', 'sg': 'T', 'st': 'T'}
    -> EARNED (verdict T, grade hereditary)   carriers=['sg', 'st']
  GETTIER SIGNATURE: carriers ['jg','jt'] vs ['sg','st'], intersection = [] ✓
  note: jt was HONESTLY earned (Smith counted the coins) — one true
  witness does not launder the smuggled jg; the poisoned atom was
  load-bearing.
------------------------------------------------------------------------
CONTROL. HONEST KNOWLEDGE (no signature)
  the sheep is really there and really seen: world agrees with s1
  [honest agent = world]
    (s1 ∨ s2)   marking={'s1': 'T'}
    -> EARNED (verdict T, grade hereditary)   carriers=['s1']
  same EARNED — but carriers coincide with the world's truth-makers;
  the signature does not fire. EARNED alone does not separate
  knowledge from Gettier; EARNED + carrier-match does.
------------------------------------------------------------------------
JURISDICTION LIMIT (measured, not hidden): FAKE BARNS (1976)
  b = 'the object before me is a barn'; Henry sees a REAL barn,
  but the county is full of papier-mache facades.
  [Henry in fake-barn county]
    b   marking={'b': 'T'}
    -> EARNED (verdict T, grade hereditary)   carriers=['b']
  carriers coincide -> the signature does NOT fire, although the
  epistemologists deny Henry knowledge. The fake-barn failure is not
  misattributed warrant; it is witness sufficiency relative to an
  ENVIRONMENT — an admission-gate question (is a barn-percept a
  sufficient witness HERE?), outside this bench's jurisdiction and
  stated so. Cf. Artemov's resolution via justification identity
  (RSL 2008 §4); ZTL's admission-gate reading is complementary.
========================================================================
E36 GREEN — two Gettier cases carry the signature (disjoint
carriers, smuggled ground); the control and the fake-barn limit do
not; the passport office finds zero paradoxes: Gettier is an
admission defect, and the ancient 'justified' is the credit ZTL
refuses.
```

Appendix B. A pocket glossary

- **Earned (T / EARNED)** — a completed verification has been produced; no path of verifications can revoke the verdict (hereditary).
- **Dirty (Z)** — not ours: its owner may change it between two reads, so each occurrence is read on its own; a verification cleans it. The logic never counts it as true. A verdict that rides a dirty link is **ON CREDIT**. (v1.0 glossed Z "unverified".)
- **Carriers** — the verified atoms the disposition actually rides on: remove one and the verdict changes.
- **The Gettier signature** — EARNED at both ends, a smuggled atom inside, carriers disjoint.
- **The admission gate** — the entry rule: a T without a sufficient witness is only Z.
- **An epoch event** — the expulsion of an admitted lie; distinguished from a verification tick by the machinery itself (E_TL_GROUND).